

Solution Requirements

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely access the application and upload voice recordings for evaluation.	High
2	Authorization levels	Student users can record, upload, view results, download reports, and access evaluation history.	High

3	External interfaces	Integrates Whisper Speech-to-Text, Gemini AI API, Sentence Transformers, SQLite Database, and PDF Generator.	High
4	Transactions processing	Accepts voice input, converts speech to text, evaluates semantic similarity, generates AI feedback, and stores results.	High
5	Reporting	Generates transcript, similarity score, AI feedback, dashboard analytics, history, and downloadable PDF reports.	High
6	Business rules, etc.	Only valid audio files are processed. Evaluation is performed only after successful transcription. Results are automatically stored.	Medium
7	Compliance to laws or regulations	User data is securely stored and used only for educational purposes while maintaining privacy.	Medium
8	<i>Other</i>	Supports multiple concept evaluations and maintains complete evaluation history.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	System should quickly process uploaded audio and generate feedback.	Evaluation completed within 10–15 seconds.
2	Scalability	System should support multiple users and future cloud deployment.	Easily scalable with additional users.
3	Security & Data Privacy	Protect uploaded audio, transcripts, and evaluation results.	Secure storage with restricted access.
4	Reliability & Availability	System should operate consistently without failures.	99% availability during normal usage.
6	<i>Other</i>	Maintainability and portability across different operating systems.	Modular codebase with easy deployment.